

RESEARCH ARTICLE

Enhancing Segmentation: A Comparative Study of Clustering Methods

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ABSTRACT With the increasing complexity of consumer preferences and behaviors, businesses face challenges to capture the dynamic nature of online consumer behavior, highlighting the need for advanced approaches. This study aims to enhance customer segmentation in e-marketing by analyzing and comparing various machine learning-based clustering methods, with a particular focus on unsupervised clustering techniques for predicting Customer Lifetime Value (CLV). While prior research has utilized unsupervised clustering for customer segmentation, this current study uniquely integrates K-Means++ with other clustering techniques to enhance segmentation accuracy and gain deeper insights into consumer behavior. This study adopts a structured, unsupervised clustering approach, enabling natural customer groupings without predefined labels, which is particularly suitable for customer segmentation in scenarios with limited labeled data. Several clustering techniques are investigated, including K-Means, K-Medoids, Agglomerative Clustering, DBSCAN, Fuzzy C-Means, K-Means++, Mini Batch K-Means, Mean Shift, and Gaussian Mixture Models (GMM). K-Means++ demonstrated superior performance in segmentation accuracy, outperforming other techniques under various conditions. Performance is evaluated using key metrics such as the Silhouette Score and Davies-Bouldin Index. Utilizing Kaggle datasets, the analysis follows a comprehensive preprocessing protocol comprising RFM (Recency, Frequency, Monetary) analysis, outlier removal, and data normalization to ensure data integrity and facilitate systematic identification of distinct consumer segments. This research highlights the potential and significance of machine learning in refining customer segmentation processes within e-marketing, ultimately aiding businesses in optimizing their marketing effectiveness and strategic planning. While focusing primarily on a limited selection of clustering methods, the study underscores the necessity for ongoing exploration in the realm of consumer segmentation. By utilizing advanced clustering methods such as K-Means++, businesses can enhance the marketing efforts to succeed in the competitive e-marketing landscape. Unlike previous studies that often relied on traditional techniques, which may not fully capture the complexities of consumer behavior, this study introduces a comprehensive approach that leverages multiple clustering methods to gain deeper insights into consumer behavior. Additionally, considering the study limitations, further research could explore additional clustering techniques, refine predictive modeling approaches and investigate the generalizability of findings to industries beyond e-marketing.

INDEX TERMS Customer segmentation, clustering, RFM, K-Means++, K-Medoids, CLV prediction.

I. INTRODUCTION

Our lives are growing more and more convenient as science and technology improve and spread globally. The recent

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Covid19 virus pandemic has made the Internet a necessity for everyday life. During that period of isolation, people used the internet for things like studying, working, and shopping [1]. As a result, people become increasingly reliant on the Internet. Today, a lot of companies have switched to e-marketing, and as more people shop online, companies face

more competition. E-marketing is a method of purchasing or offering goods online. Customers can make purchases with a single click from anywhere in the world or at any time without leaving home [2]. This increases the popularity of e-marketing around the world because of how convenient e-marketing is.

Due to the existence of many competitors in the market, each company began to find ways to enhance performance, including focusing on customer segmentation techniques. Customer segmentation is a marketing tool that divides customers into groups. In a clear strategic business model and a specific market, the company groups customers according to the customers' attributes, behaviors, needs, preferences, consumer psychology, or other characteristics, and then uses different marketing strategies for each group of customers.

This study aims to explore the field of customer segmentation in the e-marketing sector with the clustering method and business analytics techniques. By utilizing clustering techniques, the clustering method can classify the customers in different segments efficiently based on customers' behaviors and preferences. Utilizing business analytics techniques enables the company to make better marketing decisions that allows the companies to increase profits. This study will employ several clustering techniques, including K-Means, K-Means++, Mean Shift, Gaussian Mixture Model (GMM) and K-Medoids, to effectively segment customers based on behaviors and preferences.

The significance of this study lies in the potential to provide e-marketing professionals with knowledge and insights that can be referenced to improve the company's marketing efficiency and improve customer satisfaction. By implementing customer segmentation techniques, businesses can optimize the marketing strategies and provide personalized experiences to target customers. In addition, the study's findings could contribute to the broader e-marketing research field by explaining the effectiveness and utility of clustering methods for customer segmentation.

Furthermore, through this study, e-marketing professionals can gain a comprehensive understanding of the benefits, challenges, and practical applications of using clustering methods for customer segmentation in the field of e-marketing. By examining real-world data and applying advanced analytics, the study aims to generate actionable recommendations that businesses can implement to maximize the impact of the marketing efforts and foster long-term customer relationships.

Businesses nowadays must deal with a wide range of client desires and preferences to develop efficient marketing strategies in the highly competitive e-marketing sector [3]. Businesses would struggle to distinguish between various clients and appropriate strategies without efficient customer segmentation. Failure to comprehend the audience will prevent companies from increasing revenue. So, understanding the customers' desires and preferences is crucial for businesses.

However, the issue is that the e-marketing sector lacks a structured plan for consumer segmentation. Traditional segmentation techniques frequently rely on basic statistical data and fall short of capturing the changing patterns of consumer behavior and preferences in the internet space. Additionally, the manual segmentation method takes a lot of time and has disadvantages because the manual method misses slight differences in the client base as well as cannot work when the customer range is too big [4]. This highlights the necessity for more automated and advanced methods to improve segmentation accuracy and efficiency.

Consequently, e-marketing urgently needs an effective method of client segmentation. By utilizing data technology and algorithms, businesses can identify different consumer groups based on customers' purchasing trends, browsing habits, preferences, and other valuable data [5]. Addressing this issue will enhance the understanding of clients and enable businesses to develop marketing plans and products that are specifically aimed at boosting client fulfilment and loyalty.

The general objective of this research is to determine the advantages of consumer segmentation in e-marketing. The specific objectives of current study are: (1) To better understanding of consumer characteristics. (2) To examine and compare the clustering methods. (3) To increase the effectiveness of the business' marketing.

This study is divided into five sections: Section I provides the introduction, Section II focuses on the literature review, Section III outlines the methodology, Section IV presents the results and discussion, and Section V concludes the study.

II. LITERATURE REVIEW

A. A CUSTOMER SEGMENTATION

According to [6], Smith was the first to introduce "customer segmentation" also known as "market subdivision" concept. Customer segmentation is a method that involves classifying customers into various groups based on similarities in characteristics or behaviors. This approach helps businesses gain a deeper understanding of the customers and allows businesses to customize strategies and offers to meet the specific needs and preferences of each customer segment [7]. By targeting these segments with marketing efforts, businesses can increase customer satisfaction, improve marketing efficiency, and cultivate stronger customer relationships.

In e-marketing, customer segmentation has gained importance as businesses increasingly use digital data to target customers more precisely. Over time, segmentation has evolved from simple groupings based on demographics like age or income to advanced approaches that consider behavioral patterns and analytics. For example, e-marketers often segment customers based on age groups and spending habits, such as young adults with high purchase frequency or senior citizens with a preference for premium products. These strategies not only make marketing efforts more effective but also create a stronger connection with customers, improving satisfaction and loyalty.

In addition to this, customer segmentation can enhance customer experience and increase the profits of the business. In this study, customer segmentation is employed to tailor marketing strategies based on distinct consumer profiles, ultimately enhancing engagement and driving business success.

B. CLUSTERING

Clustering is a technique in machine learning where data points are grouped together based on similarities, allowing for the identification of patterns or relationships among the data without prior knowledge or guidance [8]. This technique helps find patterns or structures in data without knowing the outcomes beforehand. Clustering algorithms partition the data into groups, aiming to maximize similarity within groups and minimize similarity between groups. This method has various applications such as customer segmentation, image analysis, and anomaly detection. Clustering is useful for exploring data and organizing into meaningful groups. In this study, clustering is utilized to segment customers based on purchasing behaviors, allowing for more targeted marketing strategies and enhanced customer engagement.

C. K-MEANS, K-MEDOIDS, AGGLOMERATIVE, DBSCAN, AND FUZZY C-MEANS CLUSTERING METHODS

Customer segmentation plays a vital role in e-marketing as segmentation enables companies to target specific customer segments and improve the marketing strategies accordingly [9]. There are many types of clustering methods such as K-Means, K-Medoids, Agglomerative clustering, Density-Based Clustering Algorithms (DBSCAN) and Fuzzy C-Means. The clustering methods have been widely used to effectively segment customers. This literature review explores the findings and insights of numerous studies that use these clustering methods for customer segmentation in electronic marketing.

K-Means is a commonly used clustering method known for the effectiveness in grouping similar data points. This technique involves assigning data points to the nearest cluster center and iteratively adjusting the centers to minimize the differences within each cluster. By doing so, K-Means aims to create clusters that contain data points with similar characteristics, allowing for meaningful grouping and analysis of the data. This method is commonly used for analyzing data and uncovering patterns in a straightforward manner. K-Means clustering has become widely used in e-marketing for customer segmentation. In the study by Mufarroha et al. [10], researchers found that the optimal number of clusters using K-Means and K-Medoids clustering techniques was 4 and 6 respectively, showing that the different methods have identified different patterns within the data. However, noting that K-Means clustering is not suitable for datasets with outliers is important. Nonetheless, K-Means clustering has been shown to be efficient in marketing.

Besides that, the combination of hierarchical clustering with K-Means clustering has been suggested as a method to

identify high-potential customers and customize marketing strategies [3]. However, this approach primarily focuses on customer behavior and may not fully capture other relevant variables. Nevertheless, the integration of hierarchical and K-Means clustering techniques shows promise in enhancing marketing strategies.

In the other study by Dedi et al. [11], researchers employed K-Means clustering to determine the best cluster for targeting promotional activities towards loyal customers. Researchers acknowledge that this approach only considers some criteria, which may lead to inaccuracies. Nonetheless, the value of K-Means clustering lies in identifying customers that contribute significantly to a company's profits. Other than that, the effectiveness of K-Means clustering in identifying profit-driving customers has been highlighted [12]. However, the sensitivity of the K-Means algorithm to initialization is a known limitation. Nonetheless, companies can utilize K-Means clustering to develop effective strategies to improve profits.

K-Means clustering has been found to assist in identifying targeted customers and customers' buying patterns, enabling informed decision-making [13]. Acknowledging the challenges related to data quality, outliers, and noise is important. Nevertheless, K-Means clustering offers improved sales benefits. The researchers from study [14] have emphasized the significance of K-Means clustering in identifying different customer groups and the unique characteristics, helping companies to resolve the challenge of customer group classification.

Using K-Means clustering, researchers from study [15] categorized e-marketing customers into two clusters based on the RFM values. Out of the 102 customers, 63 belong to cluster 1 while 39 belong to cluster 2. This indicates that cluster 1 has a higher average RFM value compared to cluster 2. The study [15] noted that the initial starting points of the clusters as well as the number of clusters used can affect the results. Nonetheless, K-Means clustering enables companies to make the right decisions regarding marketing strategies. The identification of denser clusters of users through K-Means clustering has been linked to enhanced customer experience and increased business profits [16]. The study's results from [16] reveal that cluster 1 contains a larger number of users, indicating a higher concentration of users in that specific cluster. The limitation of the initial position of cluster centroids in K-Means clustering is acknowledged. Because K-Means is sensitive to the initial position of the cluster centroid and results in various cluster assignments depending on the initial location of the centroids. However, clustering can help companies to enhance customer experience and increase business profits. In this current study, K-Means clustering is applied to effectively segment e-marketing customers based on RFM values, providing actionable insights for tailoring marketing strategies and improving overall customer engagement.

The results of study [17] indicate that the combined weighted clustering method yields significantly improved

clustering outcomes for e-marketing customers compared to traditional methods. These findings demonstrate the effectiveness of utilizing K-Means clustering in improving the accuracy of customer segmentation. By employing this approach, businesses can develop more targeted marketing strategies, optimize resource allocation, and foster stronger customer loyalty. The researchers from study [18] have successfully demonstrated the value of K-Means clustering in achieving these outcomes, emphasizing the potential for driving better results in e-marketing initiatives. Determining suitable segmentation scopes in K-Means clustering presents challenges that require careful consideration of numerous factors. Nonetheless, companies can understand the product type and payment method that is popular in the company and enables to retain customers by upgrading the quality of services and products, improving communication, and gaining a better understanding of customers. In this current study, the application of clustering techniques allowed for a more detailed understanding of customer behavior, facilitating the development of personalized marketing strategies.

Another study [19] by these researchers shows that using K-Means clustering can classify customers into three distinct clusters which are Silver, Gold, and Platinum. While this segmentation has been shown to help improve service quality, companies can provide different services based on different categories of clusters. The limitation of the study [19] is that the analysis is based on data collected within a limited two-year period only. The value gained from this clustering technique is the potential to increase sales and profits. The study by Saini et al. [20] found that K-Means clustering enables firms to focus on specific customers, thereby enhancing customer experience during the product purchase process. However, a limitation of K-Means clustering is that clusters can have different shapes and sizes, which may reduce the effectiveness. However, the overall value of implementing K-Means clustering is to maximize firm revenue.

The researchers from study [21] highlighted the importance of K-Means clustering in evaluating cluster models and determining customer distributions. The impact of this approach is especially important for digital marketing companies and businesses looking to refine the strategies and achieve better results. Furthermore, other researchers from study [9] also highlighted K-Means clustering can be easier to identify the customer characteristics, thereby improving customer relationships and increasing firm revenue.

However, the study by Bhatia et al. [22] pointed out the limitations of the K-Means clustering method, as the method does not consistently produce practical and beneficial results. Nonetheless, the result will show the various categories of clusters with distinct colors which can help companies easily differentiate the customer category and improve the marketing plan based on different clusters. When successfully applied, K-Means clustering enables companies to formulate better marketing strategies and increase sales and revenue of the company. Furthermore, these two studies by Arul et al. [23] and Gankidi et al. [24] utilize the K-Means

clustering algorithm to identify customer behavior and mind-set, enabling companies to improve products, technology, and increase sales.

Besides K-Means, other clustering techniques have also proven worth in customer analysis. The researchers from study [25] use Fuzzy C-Means clustering to determine which customer clusters exhibit loyalty to a particular product. The value of this approach lies in the ability to target marketing efforts more effectively, saving time and resources. Similarly, the study by Gopal and Jacob [26] employs Agglomerative clustering to enable firms to gain a deeper understanding of customer attitudes and improve customer satisfaction. While this technique has limitations in terms of application accuracy and other performance metrics, still provides valuable insights that can help effectively solve business problems.

D. K-MEANS++ AND MINI BATCH K-MEANS

Customer analytics plays a vital role in understanding and meeting the needs of different customer segments, enabling businesses to formulate effective marketing strategies and increase profitability. Therefore, clustering algorithms are important to enterprises, such as K-Means++ and Mini Batch K-Means, which have been widely used in customer analysis to group customers according to the similarity and discover valuable insights. Many researchers have also done research on these algorithms to explore the effectiveness of these clustering techniques and the impact on customer segmentation in various situations [5], [27], [28]. Overall, these algorithms provide businesses with valuable tools to enhance the customer segmentation efforts and improve overall marketing effectiveness.

The K-Means++ algorithm has been studied in customer analysis, and researchers [5] found that this method outperformed other clustering methods. K-Means++ provides greater separation between clusters and higher closeness within clusters, making an effective tool for customer segmentation. However, this approach should consider that customer value can vary significantly within the same industry due to factors such as product type, customer preferences, and market conditions. Despite this limitation, using the K-Means++ algorithm can guide companies to implement differentiated marketing strategies to increase profits.

Another clustering technique called Mini Batch K-Means has been applied to customer analysis, especially in the sector of electronic marketing. The researchers from study [27] demonstrated that Mini Batch K-Means can effectively divide e-marketing customers into meaningful segments and reveal the unique characteristics. However, both traditional K-Means clustering and Mini Batch K-Means may encounter challenges when dealing with large datasets, such as scalability and performance issues. Nevertheless, Mini Batch K-Means can help companies to improve the marketing strategy.

Furthermore, the researcher from study [28] compared Mini Batch K-Means with other existing models, including

K-Means, Agglomerative Clustering, and Mean-Shift, and found that Mini Batch K-Means produced lesser errors in clustering compared to existing models. However, one limitation of this approach is that difficult to deploy permanently, as customer habits and buying patterns change over time. Nonetheless, implementing Mini Batch K-Means may lead to higher earnings through improved marketing efforts.

E. COMPARISON BETWEEN CLUSTERING METHODS

This section presents findings from several studies, highlighting the results, limitations, and values of each study. Various clustering methods, including K-Means, DBSCAN, Agglomerative Clustering, Mean-Shift Clustering, K-Medoids, Fuzzy C-Means, Spectral Clustering, Gaussian Mixture Model-based clustering, and Particle Swarm Clustering, have been compared across these studies [4], [8], [29], [30] to determine the effectiveness in customer segmentation for e-marketing.

The researcher from this study [7] compared the K-Means and DBSCAN algorithms and indicated that DBSCAN outperformed K-Means in identifying customers who exhibited distinct spending behaviors, setting apart as a more effective approach. However, the study did not include necessary pre-processing steps for data preparation. Despite this limitation, DBSCAN was considered useful in detecting potential customers. The other researcher [29] compared K-Means Clustering, Agglomerative Clustering, Mean-Shift Clustering, and DBSCAN Clustering. The study concluded that DBSCAN and Mean-Shift performed better than K-Means and Agglomerative in identifying meaningful clusters. However, K-Means and Agglomerative struggled to find meaningful data. The authors [29] emphasized that DBSCAN and Mean-Shift could significantly assist marketing teams in personalizing market campaigns.

Besides that, the study by Maulina et al. [4] compared K-Means, K-Medoids, and Fuzzy C-Means clustering methods. The result determined that K-Means was the most effective algorithm for cluster analysis. However, the authors noted that the characteristics of company customers in business-to-business (B2B) settings, which refer to transactions and relationships between businesses rather than between businesses and consumers, may require further improvement for accurate segmentation. This highlights the value of using K-Means to develop differentiated strategies for specific customer groups. Apart from this, the other researcher [31] also compared K-Means, Agglomerative, and Mean-Shift clustering methods, concluding that K-Means and Agglomerative performed better in clustering the data compared to Mean-Shift. However, the authors acknowledged that K-Means clustering might not fully capture complex relationships between variables, resulting in poor grouping outcomes. This underscores the importance of selecting the most suitable clustering algorithm based on the dataset characteristics.

Furthermore, the researchers from study [32] compared K-Means and K-Medoids methods and concluded that K-Means performed better. The authors noted that small

variations within customer segments might be challenging to detect. Nevertheless, the study [32] emphasized the use of K-Means for supporting customer relationship management. The other researcher [8] compared K-Means, Agglomerative Clustering, Spectral Clustering, Gaussian Mixture Model-based clustering, and DBSCAN Clustering methods. Based on the analysis of the visualized results, the researchers concluded that K-Means clustering was the most suitable approach for the given dataset. The study from [8] further highlighted the challenges associated with predefining the number of clusters and the sensitivity of results to the initialization process. In doing so, the current study underscored the significance of leveraging K-Means clustering in enhancing marketing efficiency, facilitating budget allocation for marketing activities, and identifying emerging market potentials and opportunities.

Moreover, the author [30] compared K-Means Clustering, Fuzzy C-Means, and Particle Swarm Clustering algorithms. The result found that all three methods achieved satisfactory performance. The author emphasized that the application of these methods resulted in an increase in the number of clicks on websites. Lastly, researchers from study [33] compared K-Means clustering, DBSCAN, and Affinity Propagation methods. The study observed that the sizes of the DBSCAN clusters showed significant differences, while the Affinity Propagation clusters had more balanced sizes, like the clusters generated by K-Means. The authors noted that wrong results could be produced, considering the possibility of people learning and changing habits or spending behavior. However, the study from [33] highlighted the benefit of focusing on managing each identified group effectively. In summary, the adaptability of clustering methods across different applications highlights the importance as essential tools for effective customer segmentation and marketing strategies. Table 1 presents the comparison between clustering methods across different studies.

III. METHODOLOGY

A. DATASET

The datasets that are used in this report are from the Kaggle website, which is a platform that allows users to find and publish datasets. The datasets that are chosen in this current study are retail business-related. The first dataset details the online retail transactions for a UK-based non-store that took place between 2010 and 2011 [34]. The second dataset is about the comprehensive collection of sales, customer, and product [35]. Both datasets are downloaded into a CSV file.

The first dataset has 24,700 views and 4002 downloads, consisting of 1,067,372 data and 8 columns attributes. The second dataset has 19,100 of views and 4,356 downloads, consisting of 9,994 data and 21 columns attributes.

The first dataset downloaded from Kaggle in the CSV file consists of 1,067,372 data and 8 columns, which are the following attributes: "Invoice", "Stock Code", "Description", "Quantity", "Invoice Date", "Price", "Customer"

TABLE 1. Comparison between clustering methods.

Author	Clustering Method Compared	Findings
[7]	K-Means and DBSCAN	DBSCAN outperformed K-Means in noise handling.
[29]	K-Means, Agglomerative, Mean – Shift and DBSCAN	DBSCAN and Mean-Shift performed better with varying densities than K-Means and Agglomerative.
[4]	K-Means, K Medoids and Fuzzy C Means	K-Means was the most effective for spherical clusters.
[31]	K-Means, Agglomerative and Mean Shift	K-Means and Agglomerative better for compact and hierarchical clustering than Mean-Shift.
[32]	K-Means and K Medoids	K-Means performed better.
[8]	K-Means, Agglomerative, Spectral, Gaussian Mixture Model and DBSCAN	K-Means was most suitable.
[30]	K-Means, Fuzzy C Means and Particle Swarm	All methods satisfactory.
[33]	K-means, DBSCAN and Affinity propagation	DBSCAN handles varied clusters shapes better.

TABLE 2. Features in dataset 1 without pre-processing.

#	Features	Description
1	Invoice No	The 6-digit invoice number denotes a cancellation if that code begins with the letter "C".
2	Stock Code	The 5-digit code for the product.
3	Description	The product's name.
4	Quantity	The quantities of each product for every purchase.
5	Invoice Date	The invoice's date and time.
6	Unit Price	The price of the product per unit.
7	Customer ID	A 5-digit customer number.
8	Country	Each customer's country name.

and "Country". For example, one record from the first dataset shows that an invoice with ID "536365" contains the purchase of "White Hanging Heart T-Light Holder", stock code is "85123A" with a quantity of 6, each priced at £2.55. The customer is from the UK, and the invoice date is "2010-12-01". Table 2 presents the features in dataset 1 without preprocessing.

The second dataset that downloaded consists of 9,994 data and 21 columns, which are the following attributes: "Row ID", "Order ID", "Order Date", "Ship Date", "Ship Mode", "Customer ID", "Customer Name", "Segment", "Country", "City", "State", "Postal Code", "Region", "Product ID", "Category", "Sub-Category", "Product Name", "Sales", "Quantity", "Discount" and "Profit". There is an additional attribute added manually, which is "Unit Price". This attribute calculates each product's price

TABLE 3. Features in dataset 2 without pre-processing.

#	Features	Description
1	Row ID	The number assigned to every row.
2	Order ID	A special number assigned to every order.
3	Order Date	Customer's order date.
4	Ship Date	Product shipment date.
5	Ship Mode	The ship's mode is noted and specified, including First class, Second class and Same day.
6	Customer ID	The special number assigned to every client.
7	Customer Name	Name of each client.
8	Segment	The customer segments are documented, including Consumer, Corporate and Home office.
9	Country	Each customer's country name.
10	City	Each customer's city name.
11	State	Each customer's state name.
12	Postal Code	Each customer's postal code.
13	Region	Location of every client.
14	Product ID	The distinct number assigned to every product.
15	Category	The classification of every item.
16	Sub-Category	The subcategory of every product.
17	Product Name	The name of the product.
18	Sales	The sales of every order.
19	Quantity	The number of each product for each order.
20	Discount	Every order's discount.
21	Profit	Each order's profit.
22	Unit Price	The price of the product per unit.

per unit using Excel and the formula (Sales/Quantity). For example, one record from the second dataset shows an order with the ID "CA-2013-152156" where the product "Bush Somerset Collection Bookcase", Product ID is "FUR-BO-10001798" was purchased. The order was placed by a customer named Claire Gute from the United States, with the order date being "2013-09-11" and the ship date on "2013-12-11". The order was shipped via "Second Class" with a total sale of \$261.96 for 2 units, each priced at \$130.98. The order had no discount, resulting in a profit of \$41.91. An additional "Unit Price" attribute was manually calculated as \$130.98 by dividing the sales by the quantity. Table 3 presents the features in dataset 2 without preprocessing.

B. DATA PREPROCESSING

This study uses Python to handle the missing values. After loading the dataset into Jupiter Notebook. A comprehensive check for missing values was conducted and the drop function was employed to eliminate rows containing "Nan" values. Then, the "Invoice Date" was converted to datetime format for consistent temporal analysis. Additionally, filters were applied to exclude cancelled transactions and ensure the non-negativity of "Quantity" and "Price" [10]. Figure 1 shows the results of the missing values check for dataset 1.

```

Invoice      0
StockCode    0
Description  4382
Quantity     0
InvoiceDate  0
Price        0
Customer ID  243007
Country      0
dtype: int64

```

FIGURE 1. Check missing values for dataset 1.

```

Invoice      0
StockCode    0
Description  0
Quantity     0
InvoiceDate  0
Price        0
Customer ID  0
Country      0
dtype: int64

```

FIGURE 2. Remove missing values for dataset 1.

```

Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID  0
Customer Name 0
Segment     0
Country     0
City        0
State       0
Postal Code  0
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
Quantity    0
Discount    0
Profit      0
Unit Price  0
dtype: int64

```

FIGURE 3. Check missing values for dataset 2.

Figure 2 presents the results after removing the missing values from dataset 1, while Figure 3 shows the results of the missing values check for dataset 2.

C. RFM ANALYSIS

After the data pre-processing step, RFM analysis was conducted to gain valuable insights into customer behavior. The Recency (R) score indicates how recently a customer made a purchase, with more recent purchases receiving higher scores. Frequency (F) measures how often a customer buys from the business, with more frequent buyers getting higher scores. Monetary (M) reflects how much money a customer spends, where higher spending results in a higher score. First, the consumer group should be divided according to frequency,

	CustomerId	Recency	Frequency	Monetary
0	12346.0	325	12	77556.46
1	12347.0	1	8	5633.32
2	12348.0	74	5	2019.40
3	12349.0	18	4	4428.69
4	12350.0	309	1	334.40

FIGURE 4. RFM for dataset 1.

	CustomerId	Recency	Frequency	Monetary
0	AA-10315	184	5	5563.560
1	AA-10375	19	9	1056.390
2	AA-10480	259	4	1790.512
3	AA-10645	55	6	5086.935
4	AB-10015	415	3	886.156

FIGURE 5. RFM for dataset 2.

recency, and monetary value by conducting RFM analysis on the dataset. Secondly, RFM scores should be assigned to each customer based on the behavior, such as recent purchase date, number of purchases made and total spending. Thirdly, customers should be clustered into distinct groups using the RFM scores to identify different segments based on the engagement and value to the business. Finally, the characteristics and behaviors of each RFM segment should be analyzed to gain insight into customer preferences and identify opportunities for targeted marketing strategies. Figure 4 and Figure 5 below show the RFM results for dataset 1 and dataset 2.

D. OUTLIER REMOVAL AND NORMALIZATION

In this section, a robust function has been developed to identify and eliminate outliers based on the Interquartile Range (IQR). The function is applied strategically to the “Recency”, “Frequency” and “Monetary” columns within the RFM dataset, ensuring the removal of data points that may skew the analysis. Outliers are defined as data points that fall outside the range of the first quartile (Q1) minus 1.5 times the IQR, or the third quartile (Q3) plus 1.5 times the IQR. The Interquartile Range (IQR) method provides a reliable mechanism for detecting outliers, enhancing the overall integrity of the RFM dataset [10].

Additionally, the dataset undergoes min-max normalization, this is a crucial step in standardizing the features which are Recency, Frequency, and Monetary within a consistent range of [0,1]. This normalization process is instrumental in ensuring that each feature contributes proportionally to the subsequent analyses, preventing any undue influence due to differing scales.

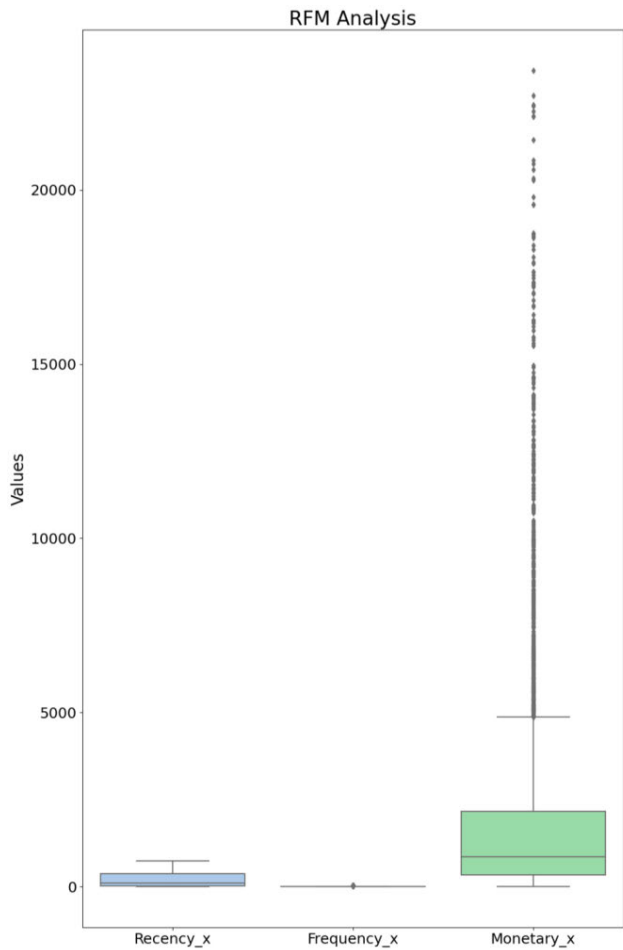


FIGURE 6. Boxplot after outlier removed for dataset 1.

By systematically removing outliers and normalizing the dataset, the analysis is poised to deliver more accurate and meaningful results, setting the stage for subsequent stages of the study. This meticulous approach enhances the reliability and interpretability of RFM data, providing a solid foundation for the ensuing analyses and insights.

Figure 6 illustrates a boxplot for Dataset 1 after outlier removal. This visualization provides a clear perspective on the distribution of Recency, Frequency, and Monetary values. The central box represents the main data points, with a line indicating the median. Removing the outliers makes the plot more compact, simplifying the interpretation of the data’s range and spread. Overall, this boxplot enhances understanding of the dataset and prepares for the next analysis steps.

Figure 7 shows a similar boxplot for Dataset 2, also reflecting the removal of outliers. This figure highlights the distribution of Recency, Frequency, and Monetary metrics. The elimination of outliers results in a more concentrated grouping of data points, suggesting that some extreme values may have distorted the results. This clearer representation boosts confidence in the dataset and paves the way for more accurate insights in future analyses.

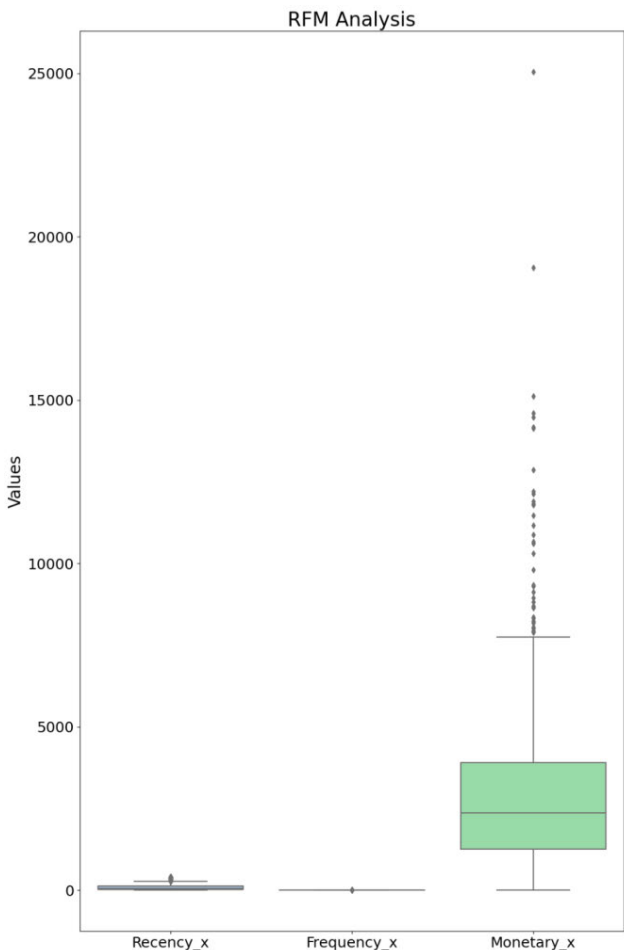


FIGURE 7. Boxplot after outlier removed for dataset 2.

	Recency_normalized	Frequency_normalized	Monetary_normalized
0	0.001355	0.166667	0.240542
1	0.100271	0.095238	0.086228
2	0.024390	0.071429	0.189104
3	0.418699	0.000000	0.014279
4	0.506775	0.000000	0.012850
...	...	...	...
5772	0.004065	0.500000	0.116855
5773	0.584011	0.000000	0.019714
5774	0.894309	0.000000	0.018233
5775	0.644986	0.023810	0.055357
5776	0.056911	0.142857	0.178613

FIGURE 8. Min-max normalization for dataset 1.

Figure 8 presents the results of min-max normalization for Dataset 1. This process adjusts the Recency, Frequency, and Monetary values to a scale between 0 and 1. By standardizing the data, all metrics can be compared more easily. This step is vital for ensuring accurate analyses and interpretations, setting a solid foundation for the next phases of the study.

Figure 9 displays the results of min-max normalization for Dataset 2. Like Figure 8, this visualization shows how the Recency, Frequency, and Monetary values have been adjusted to fall within the 0 to 1 range. This normalization eliminates any differences in scale among the features, ensuring each contributes equally to future analyses. This consistent

	Recency_normalized	Frequency_normalized	Monetary_normalized
0	0.448780	0.2500	0.222010
1	0.046341	0.5000	0.041998
2	0.631707	0.1875	0.071318
3	0.134146	0.3125	0.202974
4	0.131707	0.4375	0.309558
...	...	...	...
714	0.104878	0.6250	0.094648
715	0.009756	0.2500	0.217648
716	0.021951	0.4375	0.268214
717	0.131707	0.7500	0.320345
718	0.492683	0.2500	0.059474

FIGURE 9. Min-max normalization for dataset 2.

approach improves the reliability of the RFM data, facilitating the drawing of meaningful conclusions.

In conclusion, the boxplots provide clearer distributions after outlier removal, while min-max normalization standardizes the metrics. These processes improve the accuracy and reliability of the upcoming analyses, establishing a solid foundation for making informed decisions based on the RFM model.

E. ELBOW METHOD

The optimal number of clusters (K) in the clustering algorithms K-Means and K-Medoids can be determined graphically using the Elbow Method. Within-cluster-sum-of-square (WCSS) values are used in this method. These values are the sum of squared distances between data points and the cluster centers to which data have been assigned. The link between the corresponding WCSS values on the y-axis and the number of clusters (K) on the x-axis is represented by the elbow graph. When additional cluster addition does not significantly reduce WCSS, the graph displays an “elbow,” or apparent bend, and this is the ideal K value [10]. This point represents a balance between avoiding unnecessary complexity and minimizing intra-cluster distance.

Figure 10 illustrates the results of the Elbow Method applied to Dataset 1, displaying the relationship between the number of clusters (K) and inertia. The plot includes values for K ranging from 2 to 10, with inertia plotted on the y-axis. Each point on the curve represents the inertia calculated for a specific number of clusters. Based on the analysis, a value of K = 4 appears optimal, balancing simplicity and accuracy in segmenting the data for K-Means clustering in Dataset 1.

Figure 11 presents the results of the Elbow Method applied to Dataset 2, illustrating the relationship between the number of clusters (K) and inertia. Similar to Dataset 1, the analysis reveals that K = 4 is also optimal for this dataset, indicating a consistent trend in cluster segmentation across both datasets.

Figure 12 displays the results of the Elbow Method applied to Dataset 1 using K-Medoids, illustrating the relationship between the number of clusters (K) and inertia. The analysis shows values for K ranging from 2 to 10, with inertia plotted on the y-axis. Each point on the curve represents the inertia calculated for the corresponding number of clusters. Based on this analysis, a value of K = 6 emerges as optimal, suggesting an effective balance for segmenting the data in Dataset 1 using the K-Medoids algorithm.

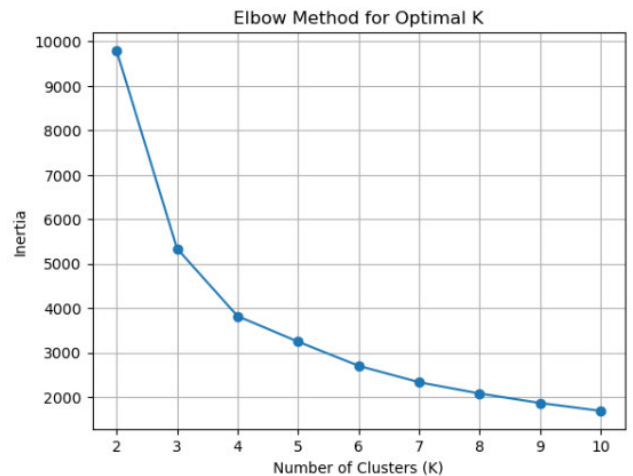


FIGURE 10. Elbow method for K-Means in dataset 1.

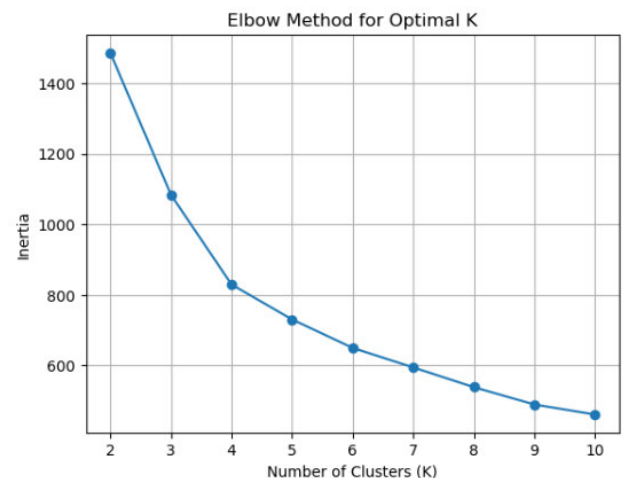


FIGURE 11. Elbow method for K-Means in dataset 2.

Figure 13 illustrates the results of the Elbow Method applied to Dataset 2 using K-Medoids, depicting the relationship between the number of clusters (K) and inertia. The analysis indicates that K = 7 is the optimal choice for this dataset, suggesting an effective balance for segmenting the data with the K-Medoids algorithm.

F. SILHOUETTE SCORE

One measure for evaluating a clustering technique's quality is the Silhouette Score. The degree of cluster separation is represented by a scale ranging from -1 to 1. A higher Silhouette Score indicates more defined clusters. Each data point's average distance from other points in the same cluster (a) and from the closest cluster to which the point does not belong (b) are considered when calculating the score for that point. A cluster's Silhouette Score is calculated as $(b - a) / \max(a, b)$. The average of these ratings over all data points is the overall Silhouette Score. Higher Silhouette Scores indicate better cluster placements.

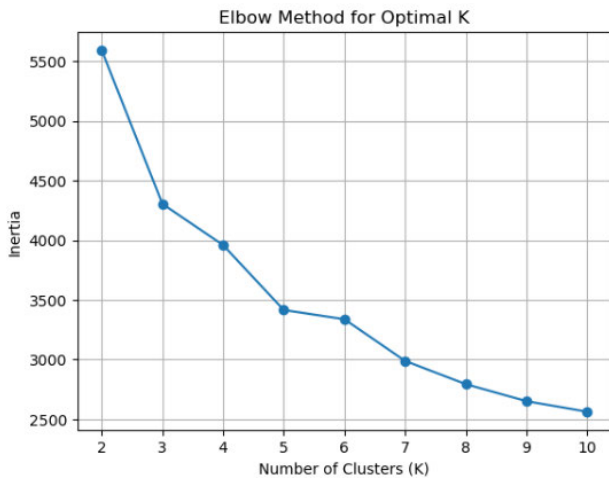


FIGURE 12. Elbow method for K-Medoids in dataset 1.

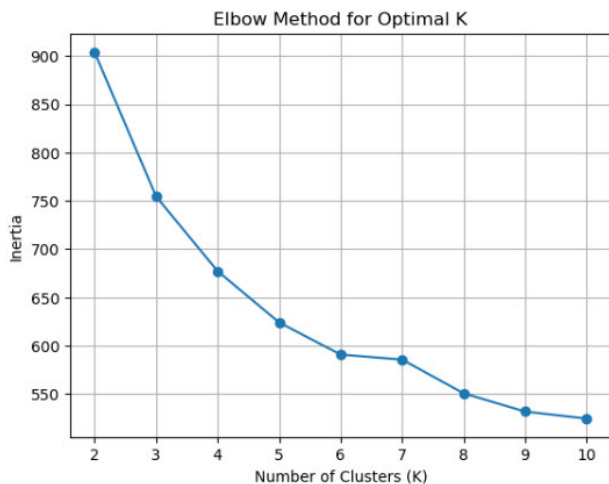


FIGURE 13. Elbow method for K-Medoids in dataset 2.

G. DAVIES-BOULDIN INDEX

Another measure for evaluating the quality of clustering is the Davies-Bouldin Index. This method evaluates how close together and apart clusters are. A lower Davies-Bouldin Index indicates better grouping. The average similarity between each cluster and the most similar cluster is found using the index. The Davies-Bouldin Index is the average of these cluster similarity values across all clusters. Like the Silhouette Score, a lower Davies-Bouldin Index signifies better-defined clusters with clear boundaries.

H. RESULTS OF SILHOUETTE SCORE AND DAVIES-BOULDIN INDEX FOR EACH CLUSTERING METHOD IN DATASET 1

Machine learning is an essential tool for data analysis, where algorithms learn from data to identify patterns and make decisions. In this study, clustering algorithms are used to segment customer data based on features such as purchase

TABLE 4. Results of silhouette score and Davies Bouldin index for k-means++.

K values	Silhouette score	Davies-Bouldin Index
4	0.501173357511503	0.7480599967897325
5	0.44757248843086334	0.7734061204265055
6	0.43823598921130397	0.8696763098199681
7	0.4411712422909567	0.8290848086022972
8	0.4013454517543362	0.8861083716730813
9	0.3989369449360252	0.884614702001465
10	0.4010285509524828	0.8961053234469419

behavior and product preferences. Different clustering methods, including K-Means++, K-Medoids, Gaussian Mixture Model (GMM) and Mean Shift are evaluated using performance metrics like the Silhouette Score and Davies-Bouldin Index.

Based on the presented tables for each clustering method, prioritizing higher Silhouette Scores and lower Davies-Bouldin Index values indicates favorable choices. Consequently, the identified optimal K values are 4 for K-Means++, 7 for K-Medoids, 8 for Gaussian Mixture Model (GMM) and 5 for Mean Shift. These specific K values are deemed suitable for achieving well-defined and compact clusters, aligning with the objective of effective customer segmentation in the context of e-marketing.

The table gives a thorough summary of the Davies-Bouldin Index and Silhouette Score for the K-Means++ clustering algorithm for a range of K values. The Davies-Bouldin Index and Silhouette Score values that correlate to each K value represent the quality of clustering at various levels of cluster granularity.

Table 4 presents the performance metrics for K-Means++ across different cluster values (K). The Silhouette Score indicates how well-defined the clusters are, with higher values suggesting better clustering. The best score of 0.501 occurs at K = 4, showing that this number of clusters provides clear separation. The scores decrease with more clusters, with values dropping to 0.401 for K = 8, 9 and 10. The Davies-Bouldin Index reflects the quality of clustering, with lower values indicating better separation between clusters. The lowest value of 0.748 is also at K = 4, suggesting effective clustering. As the k values increases, the index rises, reaching 0.896 at K = 10, indicating poorer separation. In summary, K = 4 offers the best clustering performance and balancing.

In the following subsection, an analysis of the K-Medoids clustering method is presented. The exploration delves into the Silhouette Score and Davies-Bouldin Index results across various K values, offering insights into the clustering efficacy and granularity achieved through the K-Medoids algorithm.

TABLE 5. Results of silhouette score and Davies Bouldin index for k-medoids.

K values	Silhouette score	Davies-Bouldin Index
6	0.3608054455967053	0.9689007728685867
7	0.3894031467427748	0.9561120524949739
8	0.3421329904395597	0.9088300356471752
9	0.3261422149378104	0.9767307443529624
10	0.3271557447116206	1.0143042032432383

TABLE 6. Results of silhouette score and Davies Bouldin index for gaussian mixture model.

K values	Silhouette score	Davies-Bouldin Index
8	0.04518632485982805	2.6393739309348283
9	0.00839795374184145	3.1060467504571374
10	-0.0027898642136081663	2.769921993081655

Table 5 summarizes the performance metrics for K-Medoids across different values of K. The highest Silhouette Score of 0.389 occurs at $K = 7$, indicating the best cluster separation. Although $K = 8$ has the lowest Davies-Bouldin Index at 0.908, $K = 7$ is selected as the optimal balance, providing clearer clustering while ensuring suitable separation.

In the subsequent subsection, the analysis shifts to the Gaussian Mixture Model (GMM) for clustering. The evaluation encompasses the examination of Silhouette Score and Davies-Bouldin Index results across different configurations of the GMM, shedding light on the effectiveness in capturing underlying patterns and structures within the dataset.

Table 6 shows the performance metrics for the Gaussian Mixture Model at different K values. The highest Silhouette Score of 0.045 occurs at $K = 8$, indicating the best cluster separation. The Davies-Bouldin Index also reaches the lowest point at $K = 8$ with a value of 2.639, suggesting clearer clusters. In comparison, both $K = 9$ and $K = 10$ demonstrate weaker separation, making $K = 8$ the most suitable choice for this model.

In this subsection, attention is directed towards Mean Shift clustering analysis. Mean Shift is different because this clustering can inherently detect the number of clusters based on the density of data points. Mean Shift offers a non-parametric approach to clustering, effectively identifying dense regions of data points in feature space. The evaluation entails examining Silhouette Score and Davies-Bouldin Index results across varying bandwidth parameters, elucidating Mean Shift's ability to adaptively determine cluster centers and capture data distribution characteristics.

TABLE 7. Results of silhouette score and davies bouldin index for mean shift.

K values	Silhouette score	Davies-Bouldin Index
5	0.4026639739639286	1.3553122775428927

Table 7 displays the results for the Mean Shift clustering method at $K = 5$. A Silhouette Score of 0.403 indicates decent cluster separation, while a Davies-Bouldin Index of 1.355 suggests moderate compactness. These results imply that $K = 5$ provides a good balance between cluster separation and compactness.

I. CLUSTERING USED

Informed by the literature review in Chapter 2, various partition-based and density-based clustering algorithms have been employed in previous studies. For this study, the chosen clustering approaches include K-Means, K-Means++, K-Medoids, Gaussian Mixture Models (GMM) and Mean Shift.

K-Means is a widely used partition-based clustering algorithm, is selected for simplicity and effectiveness [10]. Using this approach, data points are divided into K clusters, and each point is allocated to the cluster with the closest mean. The iterative refinement process continues until a convergence criterion is met.

K-Means++ is an improvement over the standard K-Means algorithm in terms of initializing cluster centroids. By employing a smarter initialization strategy, K-Means++ often converges faster and provides more robust clustering results [5].

Table 8 presents the K-Means++ clustering results for Dataset 1, categorizing customers into four distinct groups based on their Recency, Frequency, and Monetary values. Cluster 0 includes customers with low recency but high frequency and monetary values, indicating that this group tends to make purchases frequently and demonstrates strong loyalty. Cluster 1 consists of customers who have made recent purchases but do not engage often, as shown by the high recency and low frequency and monetary scores. Cluster 2 is made up of customers with low values across all metrics, suggesting that this group is inactive and might require targeted strategies to encourage a return. Cluster 3 features low recency, high frequency, and high monetary values, representing another segment of loyal customers who consistently make purchases.

Table 9 displays the K-Means++ results for Dataset 2, which also includes four clusters. Cluster 0 stands out with high recency, frequency, and monetary values, indicating highly engaged customers who frequently purchase and spend well. Cluster 1 features customers with high recency but low frequency and monetary values, suggesting that while these customers have recently made purchases, but are

TABLE 8. Results of K-Means++ cluster for dataset 1.

Cluster	Recency	Frequency	Monetary
0	Low	High	High
1	High	Low	Low
2	Low	Low	Low
3	Low	High	High

TABLE 9. Results of K-Means++ cluster for dataset 2.

Cluster	Recency	Frequency	Monetary
0	High	High	High
1	High	Low	Low
2	Low	High	High
3	Low	Low	Low

not regular buyers. Cluster 2 highlights customers with low recency but high frequency and monetary values, showcasing a loyal group that makes consistent purchases. Lastly, Cluster 3 exhibits low value across all metrics, indicating that this group may need special marketing efforts to enhance the engagement.

K-Medoids is a partition-based clustering algorithm that, unlike K-Means, utilizes data points themselves as representatives of clusters [10]. This approach can be particularly useful in scenarios where the mean may not be a robust representative.

Table 10 shows the results of the K-Medoids clustering method applied to Dataset 1, revealing seven distinct customer clusters based on Recency, Frequency, and Monetary values. Cluster 0 consists of customers with low recency, high frequency, and high monetary values, indicating a segment of loyal customers who frequently engage in significant purchases. Cluster 1 features customers characterized by high recency, low frequency, and low monetary values, representing those who may have made recent purchases but lack ongoing engagement. Cluster 2 includes customers with high recency, high frequency, and high monetary values, suggesting a group of active buyers that should be prioritized for retention strategies. Cluster 3 exhibits high recency, low frequency, and low monetary values, indicating a segment that may need targeted marketing efforts to encourage repeat purchases. Cluster 4 also consists of customers with low recency, high frequency, and high monetary values, highlighting another group of valuable customers. Cluster 5 features customers characterized by low recency, low frequency, and low monetary values, indicating a higher risk of churn. Finally, Cluster 6 includes customers with high recency, low frequency, and low monetary values, suggesting a need for strategies to convert recent buyers into regular customers.

TABLE 10. Results of K-Medoids cluster for dataset 1.

Cluster	Recency	Frequency	Monetary
0	Low	High	High
1	High	Low	Low
2	High	High	High
3	High	Low	Low
4	Low	High	High
5	Low	Low	Low
6	High	Low	Low

Table 11 presents the K-Medoids clustering results for Dataset 2, which identifies eight customer clusters. Cluster 0 includes customers with low recency, low frequency, and high monetary values, indicating a unique segment that makes significant purchases infrequently. Cluster 1 features customers characterized by low recency, high frequency, and high monetary values, representing a loyal group of customers who engage frequently. Cluster 2 also consists of customers with low recency, high frequency, and high monetary values, suggesting a robust segment of repeat buyers. Cluster 3 consists of customers with high recency, low frequency, and low monetary values, indicating potential candidates for targeted marketing efforts to drive engagement. Cluster 4 includes customers with high recency, high frequency, and high monetary values, highlighting active customers who consistently make significant purchases. Cluster 5 features customers characterized by low recency, high frequency, and low monetary values, suggesting a group that may require strategies to increase overall spending. Cluster 6 includes customers with high recency, low frequency, and low monetary values, indicating a need for strategies to enhance engagement and retention. Finally, Cluster 7 consists of customers with low recency, low frequency, and low monetary values, suggesting a segment at risk of being inactive.

Gaussian Mixture Models (GMM) is a model-based clustering technique assuming that data points come from a combination of different Gaussian distributions. GMM is well-suited for situations where the dataset exhibits complex structures that cannot be adequately represented by simple geometric shapes [8].

Table 12 presents the results of the Gaussian Mixture Model clustering for Dataset 1, revealing eight distinct customer segments based on Recency, Frequency, and Monetary values. Cluster 0 includes customers with high recency, high frequency, and high monetary values, indicating active shoppers who frequently make significant purchases. Cluster 1 shows customers with high recency, low frequency, and low monetary values, suggesting infrequent shoppers who have recently made a purchase but may not contribute much to overall revenue. Cluster 2 is made up of customers

TABLE 11. Results of K-Medoids cluster for dataset 2.

Cluster	Recency	Frequency	Monetary
0	Low	Low	High
1	Low	High	High
2	Low	High	High
3	High	Low	Low
4	High	High	High
5	Low	High	Low
6	High	Low	Low
7	Low	Low	Low

TABLE 12. Results of gaussian mixture model cluster for dataset 1.

Cluster	Recency	Frequency	Monetary
0	High	High	High
1	High	Low	Low
2	Low	High	Low
3	Low	High	High
4	Low	High	High
5	Low	High	High
6	High	Low	Low
7	Low	Low	Low

who exhibit low recency, high frequency, and low monetary values, reflecting regular buyers who tend to make smaller purchases. Clusters 3, 4 and 5 feature customers with low recency, high frequency, and high monetary values, indicating multiple segments of loyal and valuable customers who purchase often. Cluster 6 consists of customers with high recency, low frequency, and low monetary values, pointing to those who recently shopped but do not engage frequently. Finally, Cluster 7 represents customers with low recency, low frequency, and low monetary values, signaling a group at risk of disengagement.

Table 13 shows the results of the Gaussian Mixture Model clustering for Dataset 2, identifying three distinct clusters. Cluster 0 contains customers with low recency, high frequency, and low monetary values, suggesting regular buyers who tend to spend less. Cluster 1 includes customers with high recency, low frequency, and low monetary values, indicating shoppers who have recently made a purchase but do not engage regularly. Cluster 2 highlights customers with high recency, high frequency, and high monetary values, reflecting an active and valuable segment of shoppers who frequently make significant purchases.

TABLE 13. Results of gaussian mixture model cluster for dataset 2.

Cluster	Recency	Frequency	Monetary
0	Low	High	Low
1	High	Low	Low
2	High	High	High

TABLE 14. Results of mean shift cluster for dataset 1.

Cluster	Recency	Frequency	Monetary
0	Low	Low	Low
1	High	High	High
2	Low	High	High
3	High	High	High
4	Low	High	High

Mean Shift is a density-based clustering method that identifies clusters by locating maxima in the data density function. This approach is effective in situations where clusters have irregular shapes and densities [29].

Table 14 outlines the Mean Shift clustering for Dataset 1, identifying five distinct customer groups based on Recency, Frequency, and Monetary values. Cluster 0 represents customers with low recency, low frequency, and low monetary values, indicating a segment of less engaged customers who rarely make purchases. Cluster 1 consists of highly active and valuable shoppers, characterized by high recency, high frequency, and high monetary values. Cluster 2 captures customers with low recency, high frequency, and high monetary values, suggesting a loyal group that purchases often and significantly contributes to revenue. Cluster 3 reflects similar characteristics to Cluster 1, with high recency, high frequency, and high monetary values, forming another highly engaged group. Finally, Cluster 4 displays low recency, high frequency, and high monetary values, indicating frequent and valuable shoppers.

Table 15 illustrates the Mean Shift clustering for Dataset 2, identifying seven unique clusters. Cluster 0 represents customers with low recency, low frequency, and low monetary values, indicating a less engaged group. Cluster 1 shows customers with low recency, high frequency, and high monetary values, suggesting frequent and valuable shoppers. Cluster 2 consists of customers with high recency, low frequency, and high monetary values, pointing to those who have recently made a significant purchase but do not engage frequently. Cluster 3 includes customers with high recency, high frequency, and high monetary values, indicating active and high-spending customers. Cluster 4 with high recency, low frequency, and high monetary values, forming another

TABLE 15. Results of mean shift cluster for dataset 2.

Cluster	Recency	Frequency	Monetary
0	Low	Low	Low
1	Low	High	High
2	High	Low	High
3	High	High	High
4	High	Low	High
5	Low	Low	High
6	Low	High	High

segment of recent but infrequent shoppers. Cluster 5 represents customers with low recency, low frequency, and high monetary values, suggesting a small but valuable group of infrequent buyers. Cluster 6 features customers with low recency, high frequency, and high monetary values, forming a segment of loyal and high-value customers.

The selected clustering methods are implemented after the RFM analysis and a series of preprocessing steps, ensuring the data is well-prepared for accurate and meaningful segmentation. The determination of the optimal number of clusters is conducted using the elbow method, silhouette score, and Davies-Bouldin index, providing a robust foundation for subsequent analysis. In conclusion, thorough data preparation, the use of effective clustering algorithms, and reliable evaluation methods have been essential in producing accurate customer segments, leading to improved customer engagement and greater business success.

J. CUSTOMER LIFETIME VALUE (CLV) PREDICTION

Customer Lifetime Value (CLV) prediction is a method used to estimate the long-term value and profitability of a customer for a business [3]. The prediction involves analyzing historical customer data to study the potential revenue and profitability in the future. This process includes collecting and pre-processing customer data, segmenting customers based on behavior using RFM segments and cluster method, developing predictive models, validating the models, and applying the predictions to inform business strategies [10]. Businesses can optimize overall income by using CLV prediction to assist with decisions regarding price, marketing, client retention, and resource allocation. CLV prediction helps businesses understand the value of each customer and optimize the strategies accordingly. The Mean Absolute Error calculates the average absolute difference between the expected and actual values. The model’s predictions are more accurate when the Mean Absolute Error values are lower. The average of the squares of the mistakes is measured by the Mean Squared Error. Better prediction accuracy from the model is indicated by a lower mean squared error number [3]. Based on the independent variable, the R-squared indicates what

TABLE 16. Results of random forests and gradient boosting for K-Means++ in dataset 1.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	2.3903	15.5569	2.4558	2.4212
Mean Squared Error	243.4673	831.1362	348.9028	11.9509
R-squared	0.9999	0.9998	0.9999	0.9999

proportion of the variation of the dependent variable can be expected. Consequently, greater model-data fit is indicated by larger R-squared values [36].

Moving forward, the current study implements Gradient Boosting, a potent machine learning algorithm, to enable the prediction of Customer Lifetime Value (CLV). This ensemble learning method combines weak learners into a strong predictive model and is effective in capturing complex patterns in customer behavior.

Next, applying Random Forests, another ensemble learning algorithm, known for robustness and capability to handle diverse data patterns, enhances predictive accuracy. Random Forests can provide accurate CLV predictions based on the algorithm’s ability to handle high-dimensional datasets.

Then, this current study enhances the performance of both Gradient Boosting and Random Forests models by employing Hyperparameter Tuning, which involves fine-tuning the model’s hyperparameters to enhance the performance and predictive accuracy.

This subsection focuses on the utilization of Random Forests and Gradient Boosting in conjunction with the K-Means++ clustering method. The examination delves into the predictive capabilities and model performance when applied to the clusters generated by K-Means++ across the specified datasets.

Table 16 presents the results of Random Forests and Gradient Boosting models applied to Dataset 1 using K-Means++ clustering, both before and after Hyperparameter Tuning. The performance of these models is measured by Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared values. Before tuning, Random Forests showed an MAE of 2.3903, an MSE of 243.4673, and an R-squared value of 0.9999, while Gradient Boosting had a significantly higher MAE of 15.5569 and an MSE of 831.1362, with an R-squared of 0.9998. After Hyperparameter Tuning, Random Forests showed a slight improvement in MAE to 2.4558, while the R-squared value remained constant at 0.9999, while Gradient Boosting showed substantial improvement with an MAE of 2.4212 and an MSE of 11.9509, maintaining a strong R-squared of 0.9999.

Table 17 displays the results for Dataset 2, where similar trends are observed. Before tuning, Random Forests had an

TABLE 17. Results of random forests and gradient boosting for K-Means++ in dataset 2.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	15.5362	25.3906	14.5254	1.5592
Mean Squared Error	2053.1072	2566.7146	1601.4375	5.0941
R-squared	0.9995	0.9994	0.9996	0.9999

TABLE 18. Results of random forests and gradient boosting for K-Medoids in dataset 1.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	2.3447	15.4840	2.4725	2.4222
Mean Squared Error	196.6169	800.6385	320.8613	11.9070
R-squared	0.9999	0.9998	0.9999	0.9999

MAE of 15.5362, an MSE of 2053.1072, and an R-squared of 0.9995, while Gradient Boosting produced higher errors with an MAE of 25.3906 and an MSE of 2566.7146, resulting in an R-squared of 0.9994. After Hyperparameter Tuning, Random Forests demonstrated improvements, achieving an MAE of 14.5254, an MSE of 1601.4375, and a slight enhancement in R-squared to 0.9996. Gradient Boosting showed remarkable improvement, with an MAE of 1.5592, an MSE of 5.0941, and an R-squared of 0.9999, indicating highly accurate predictions after tuning.

This subsection focuses on evaluating the performance of Random Forests and Gradient Boosting in conjunction with the K-Medoids clustering method. The analysis aims to assess the predictive accuracy and model robustness when applied to clusters generated by K-Medoids across the designated datasets.

Table 18 showcases the results of Random Forests and Gradient Boosting models on Dataset 1 using K-Medoids clustering, before and after Hyperparameter Tuning. Initially, Random Forests recorded an MAE of 2.3447, an MSE of 196.6169, and an R-squared of 0.9999. Gradient Boosting, on the other hand, had a higher MAE of 15.4840, an MSE of 800.6385, and an R-squared of 0.9998. After tuning, Random Forests experienced a slight increase in both MAE and MSE, reaching values of 2.4725 and 320.8613, while R-squared remained stable at 0.9999. Gradient Boosting experienced substantial gains, reducing the MAE to 2.4222 and MSE to 11.9070, while maintaining a strong R-squared of 0.9999.

TABLE 19. Results of random forests and gradient boosting for K-Medoids in dataset 2.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	16.5303	25.3454	15.2979	1.5649
Mean Squared Error	2349.6705	2530.7474	1874.0583	5.3597
R-squared	0.9995	0.9995	0.9996	0.9999

Table 19 presents the results for Dataset 2. Before tuning, Random Forests had an MAE of 16.5303, an MSE of 2349.6705, and an R-squared value of 0.9995. Gradient Boosting showed higher errors with an MAE of 25.3454 and an MSE of 2530.7474, resulting in an R-squared of 0.9995. After tuning, Random Forests improved with a reduced MAE of 15.2979, an MSE of 1874.0583, and an R-squared of 0.9996. Gradient Boosting achieved significant improvements, lowering the MAE to 1.5649, the MSE to 5.3597, and reaching an R-squared value of 0.9999.

Within this subsection, the emphasis is on examining the integration of Random Forests and Gradient Boosting with the Gaussian Mixture Model (GMM) clustering method. The evaluation seeks to elucidate the predictive capabilities and model adaptability within the context of GMM-generated clusters across the specified datasets [37].

Table 20 details the performance of the Random Forests and Gradient Boosting models on Dataset 1 using the Gaussian Mixture Model, comparing results before and after Hyperparameter Tuning. Initially, Random Forests achieved a Mean Absolute Error (MAE) of 2.4879, a Mean Squared Error (MSE) of 316.0545, and an R-squared value of 0.9999. In contrast, Gradient Boosting reported a higher MAE of 15.7106, an MSE of 824.4635, and an R-squared of 0.9998. After Hyperparameter Tuning, Random Forests experienced a slight increase in MAE to 2.4965 and MSE to 322.9151, while R-squared remained stable at 0.9999. Gradient Boosting demonstrated significant improvements, reducing the MAE to 2.4581 and lowering the MSE to 12.1213, with the R-squared remaining at 0.9999.

Table 21 presents the outcomes for Dataset 2, where a similar trend is observed. Before tuning, Random Forests had an MAE of 16.1810, an MSE of 2107.2269, and an R-squared of 0.9995, while Gradient Boosting produced a higher MAE of 25.3100 and an MSE of 2569.4302, yielding an R-squared of 0.9994. After Hyperparameter Tuning, Random Forests improved the MAE to 15.9164, with an MSE of 2286.6050, while the R-squared remained unchanged at 0.9995. In contrast, Gradient Boosting displayed remarkable improvement, reducing the MAE to 1.5741 and lowering the MSE to 5.5292, while achieving a higher R-squared value of 0.9999.

TABLE 20. Results of random forests and gradient boosting for gaussian mixture model in dataset 1.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	2.4879	15.7106	2.4965	2.4581
Mean Squared Error	316.0545	824.4635	322.9151	12.1213
R-squared	0.9999	0.9998	0.9999	0.9999

TABLE 21. Results of random forests and gradient boosting for gaussian mixture model in dataset 2.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	16.1810	25.3100	15.9164	1.5741
Mean Squared Error	2107.2269	2569.4302	2286.6050	5.5292
R-squared	0.9995	0.9994	0.9995	0.9999

In this subsection, attention is directed towards exploring the compatibility of Random Forests and Gradient Boosting with the Mean Shift clustering method [31]. The analysis endeavors to unveil the predictive performance and model versatility when applied to clusters derived through Mean Shift clustering across the predefined datasets.

Table 22 displays the performance metrics of Random Forests and Gradient Boosting models applied to Dataset 1 using the Gaussian Mixture Model, presenting results both before and after Hyperparameter Tuning. Initially, Random Forests recorded a Mean Absolute Error (MAE) of 2.4686, a Mean Squared Error (MSE) of 247.0451, and an R-squared value of 0.9999. In contrast, Gradient Boosting had a higher MAE of 15.7106, with an MSE of 824.4635 and an R-squared of 0.9998. After Hyperparameter Tuning, Random Forests showed a slight increase in MAE to 2.4930 and MSE to 333.5033, while the R-squared remained at 0.9999. Meanwhile, Gradient Boosting improved the MAE slightly to 2.4738 and significantly reduced the MSE to 12.4526, maintaining an R-squared of 0.9999.

Table 23 presents similar findings for Dataset 2. Before tuning, Random Forests had an MAE of 15.3888, an MSE of 2125.6122, and an R-squared value of 0.9995. Gradient Boosting reported a higher MAE of 25.5233 and an MSE of 2608.9479, with an R-squared of 0.9994. After tuning, Random Forests improved the MAE to 14.2419 and decreased the MSE to 1814.0673, while the R-squared increased to 0.9996. Conversely, Gradient Boosting achieved remarkable

TABLE 22. Results of random forests and gradient boosting for gaussian mixture model in dataset 1.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	2.4686	15.7106	2.4930	2.4738
Mean Squared Error	247.0451	824.4635	333.5033	12.4526
R-squared	0.9999	0.9998	0.9999	0.9999

TABLE 23. Results of random forests and gradient boosting for gaussian mixture model in dataset 2.

Method	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Random Forests	Gradient Boosting	Random Forests	Gradient Boosting
Mean Absolute Error	15.3888	25.5233	14.2419	1.5776
Mean Squared Error	2125.6122	2608.9479	1814.0673	5.4238
R-squared	0.9995	0.9994	0.9996	0.9999

enhancements, reducing the MAE to 1.5776 and MSE to 5.4238, resulting in a higher R-squared of 0.9999.

Based on the tables above, the Gradient Boosting method significantly outperforms Random Forests after hyperparameter tuning for predicting Customer Lifetime Value. Gradient Boosting exhibits a much lower Mean Squared Error and a higher R-squared value after hyperparameter tuning, making the better choice for predicting Customer Lifetime Value.

K. CUSTOMER LIFETIME VALUE (CLV) PREDICTION USING BEST MODEL

After completing the hyperparameter tuning process, the best-tuned Gradient Boosting model is employed for the final Customer Lifetime Value (CLV) prediction. Subsequently, an analysis is conducted to compare the predicted CLV with the Historical CLV. If the predicted CLV is less than the Historical CLV, the cluster is categorized as 'Low.' Conversely, if the predicted CLV exceeds the Historical CLV, the cluster is designated as 'High.' This step ensures that the predictive model is not only optimized for accuracy and reliability but also provides actionable insights into the relative value of each cluster.

IV. RESULTS AND DISCUSSION

A. CONCEPTUAL FRAMEWORK

The first step in the conceptual framework of the study is the preprocessing of data on selected datasets, such as datasets 1 and 2. This crucial step ensures that the dataset

is cleaned and converted to format ready for analysis. After preprocessing the data, perform RFM analysis to investigate customer behavior patterns based on currency, frequency, and recency variables. Then, to improve the consistency and quality of the data, outlier removal, and normalization procedures are utilized. Next, the Elbow method was employed to determine the optimal number of clusters for segmentation, and the clustering quality was then evaluated using the Silhouette Score and Davies-Bouldin Index. Figure 14 presents the conceptual framework for this study. The framework further explores various clustering methods, including K-Means, K-Means++, K-Medoids, GMM, and Mean Shift, aiming to identify better-defined clusters for enhanced segmentation. From the results obtained from this study, better-defined clusters were observed for specific clustering methods: K-Means++ with a Silhouette score of 0.5011 for four clusters and a Davies-Bouldin index of 0.7480; K-Medoids with seven clusters showed a Silhouette score of 0.3894 and a Davies-Bouldin index of 0.9561; GMM with eight clusters produced a Silhouette score of 0.0451 and a Davies-Bouldin index of 2.6393; and Mean Shift with five clusters showed a Silhouette score of 0.4026 and a Davies-Bouldin index of 1.3553. Customer lifetime value (CLV) prediction is carried out using Random Forests and Gradient Boosting techniques following cluster validation. To maximize model performance, hyperparameter tuning is carried out using the Randomized Search CV method. After hyperparameter optimization, the comparison output shows that the Gradient Boosting approach performs better than Random Forests, especially in predicting CLV. To allow robust customer behavior analytics and predictive modelling, this all-inclusive framework combines data preparation, RFM analysis, outlier removal, normalization, cluster validation, CLV prediction, and hyperparameter tuning.

However, certain limitations exist within this framework. Scalability could be an issue when applying these methods to large datasets, as some clustering techniques, particularly GMM and Mean Shift, may face challenges processing high-dimensional or large-scale data efficiently. The sensitivity to initial parameters also affects certain methods, particularly K-Means and K-Means++, where cluster quality can vary depending on initial cluster centroids. Lastly, the quality of data preprocessing plays a critical role in the overall performance of the framework. Inaccurate data preprocessing or poor handling of noise and outliers can lead to suboptimal clustering results and predictive accuracy.

B. COMPARISON OF CLUSTERING METHODS

In assessing the effectiveness of clustering methods, two key metrics were employed: the Silhouette Score and the Davies-Bouldin Index. The Silhouette Score, illustrating cluster cohesion and separation, exhibited a discernible range across different K values. Higher Silhouette Scores, particularly for specific K values, suggested more well-defined clusters [38]. Simultaneously, the Davies-Bouldin Index, assessing cluster

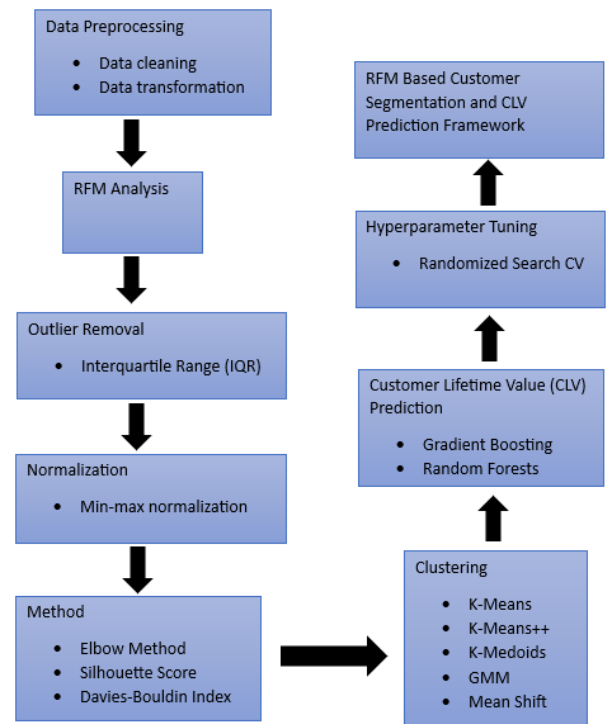


FIGURE 14. Conceptual framework for this study.

compactness and separation, displayed variability with distinct K values. Lower Davies-Bouldin Index values indicated improved clustering solutions [38]. Implementing the Elbow Method [10] and scrutinizing Silhouette Scores and Davies-Bouldin Index, potential optimal K values were identified. A comprehensive summary table detailing K, Silhouette Score, and Davies-Bouldin Index for each attempt facilitated a methodical comparison. The final selection of K involved a nuanced assessment, integrating quantitative metrics with domain expertise. This methodological approach contributed to a thorough comprehension of cluster quality, leading to the selection of a suitable number of clusters for customer segmentation.

In evaluating the performance of different clustering methods on the first dataset, the following observations were made. K-Means++ clustering method exhibits the highest Silhouette Score and a low Davies-Bouldin Index, indicating well-defined clusters and effective separation between them. K-Means++ with $K = 4$ emerges as a promising clustering approach. While K-Medoids demonstrates a decent Silhouette Score, the higher Davies-Bouldin Index suggests less distinct clusters compared to K-Means++. Nevertheless, K-Medoids still performs well in clustering. For GMM, despite having a low Silhouette Score and a high Davies-Bouldin Index, provides insights into potential challenges in achieving clear cluster definitions with this method. Mean Shift demonstrates a reasonable Silhouette Score and Davies-Bouldin Index, indicating acceptable clustering performance.

The dataset's dimensionality and noise influenced clustering performance. GMM struggled with high-dimensional

data, while Mean Shift was affected by noise, which reduced effectiveness. K-Medoids performed reasonably well, but clustering quality was less distinct compared to K-Means++ due to the higher Davies-Bouldin Index. In contrast, K-Means++ showed robustness despite these challenges, maintaining high performance and well-separated clusters. While K-Means++ provided the most consistent results, methods like GMM, Mean Shift and K-Medoids faced limitations due to noise and dimensionality.

In comparison to the benchmark paper, which used K-Means, K-Medoids, and the Elbow Method for clustering evaluation, this study extended the analysis by incorporating K-Means++, GMM and Mean Shift. While the benchmark relied on the Elbow Method for cluster selection, this study added the Silhouette Score and Davies-Bouldin Index for a more comprehensive assessment of clustering quality. The superior performance of K-Means++ demonstrated by a higher Silhouette Score and a lower Davies-Bouldin Index, aligns with known advantages, such as better cluster separation and reduced sensitivity to initialization.

Table 24 presents the results of the comparison of various clustering methods used in the analysis. Each method is evaluated based on the number of clusters (K values), the Silhouette score, and the Davies-Bouldin index. K-Means++ achieves the highest Silhouette score of 0.5012 and the lowest Davies-Bouldin index at 0.7481, indicating superior cluster separation. In contrast, the Gaussian Mixture Model (GMM) performs poorly, with a Silhouette score of 0.0452 and a high Davies-Bouldin index of 2.6394, indicating weak clustering performance. K-Medoids and Mean Shift present intermediate results, with K-Medoids yielding a Silhouette score of 0.3894 and a Davies-Bouldin index of 0.9561, while Mean Shift has a Silhouette score of 0.4027 and a Davies-Bouldin index of 1.3553. Based on this comprehensive analysis, K-Means++ with $K = 4$ stands out as the most promising clustering method. This conclusion is drawn from K-Means++'s highest Silhouette Score (0.5012) which indicates well-defined and cohesive clusters among the compared methods and a low Davies-Bouldin Index (0.7481) suggesting clear separation between clusters when compared to alternative approaches. The higher Silhouette Score of K-Means++ implies that the clusters are more cohesive and distinct, a key factor in effective segmentation. Additionally, the lower Davies-Bouldin Index reinforces the method's ability to create compact and well-separated clusters, providing further evidence of K-Means++'s superior clustering performance. In comparison to other methods, this combination of higher cohesion and better separation makes K-Means++ the most reliable and efficient choice for customer segmentation.

C. COMPARISON BETWEEN METHOD 1 [10] AND METHOD 2 (PROPOSED) FOR DATASET 1 AND DATASET 2

The comparative analysis between Method 1 [10] and Method 2 (Proposed method) across Dataset 1 and Dataset 2, employing K-Means and K-Medoids clustering methods, elucidates distinct clustering characteristics. While there

TABLE 24. Results of comparison of clustering methods.

Method	K values	Silhouette score	Davies-Bouldin index
K-Means++	4	0.5012	0.7481
K-Medoids	7	0.3894	0.9561
GMM	8	0.0452	2.6394
Mean Shift	5	0.4027	1.3553

is consistency between K-Means and K-Means++ results, slight variations occur in the output when the code is executed multiple times for K-Means clustering. For getting the best values using K-Means clustering, the trials need to repeat a few times [11]. Consequently, K-Means++, as an enhancement of the original method, emerges as the preferred choice for stability and performance reliability [5]. In the case of K-Medoids clustering, Method 2 suggests optimal K values that outperform those recommended by Method 1, as evidenced by higher Silhouette Scores and lower Davies-Bouldin Index values. The comprehensive comparison underscores the robustness of Method 2, positioning a more effective approach for cluster analysis across diverse datasets.

The performance between K-Means and K-Means++ under both Method 1 and Method 2 across Dataset 1 and Dataset 2. While the analysis meticulously examines the clustering results, underscores that although both methods exhibit comparable performance in capturing meaningful patterns within the datasets, K-Means++ demonstrates greater stability and consistency upon multiple executions of the code. This suggests that K-Means++ offers better reliability in producing consistent clustering outcomes.

Table 25 compares Method 1 (K-Means) and Method 2 (K-Means++) using Dataset 1. Both methods use 4 clusters (K), as confirmed by the Elbow method. The Silhouette Score for both methods is about 0.5012, indicating that the clusters are reasonably well-defined. This score means that items within the same cluster are similar to each other. The Davies-Bouldin Index is also the same for both methods, around 0.7481, suggesting that the clusters are well separated. Overall, Table 23 shows that both methods perform equally well in terms of clustering for Dataset 1.

Table 26 compares Method 1 (K-Means) and Method 2 (K-Means++) using Dataset 2. Again, both methods use 4 clusters, which the Elbow method confirms. For Dataset 2, both methods have the same Silhouette Score of about 0.3394, indicating that the clusters are less distinct compared to Dataset 1. This means the items in the same cluster are not as similar. The Davies-Bouldin Index is also similar, at around 0.9948, showing that the clusters are still somewhat separated but not as well as in Dataset 1. In summary, both methods yield consistent results for Dataset 2, but the clustering performance is weaker than in Dataset 1.

TABLE 25. Results of comparison of method 1 and method 2 in dataset 1.

Method	Method 1 (Authored)	Method 2 (Proposed)
	K-Means	K-Means++
K	4	4
Elbow method	4	4
Silhouette Score	0.501173357511503	0.501173357511503
Davies-Bouldin Index	0.7480599967897325	0.7480599967897325

TABLE 26. Results of comparison of method 1 and method 2 in dataset 2.

Method	Method 1 (Authored)	Method 2 (Proposed)
	K-Means	K-Means++
K	4	4
Elbow method	4	4
Silhouette Score	0.339423609882861	0.339423609882861
Davies-Bouldin Index	0.9948224668429211	0.9948224668429211

TABLE 27. Results of comparison of method 1 and method 2 in dataset 1.

Method	Method 1 (Authored)	Method 2 (Proposed)
	K-Medoids	
K	6	7
Elbow method	6	6
Silhouette Score	0.3608054455967053	0.3894031467427748
Davies-Bouldin Index	0.9689007728685867	0.9561120524949739

Next, the focus centers on the comparison of K-Medoids clustering under Method 1 and Method 2 across Dataset 1 and Dataset 2. Notably, the analysis reveals that Method 2 outperforms Method 1 for K-Medoids clustering, highlighting the effectiveness in generating clusters that accurately represent the underlying structure and characteristics of the data.

Table 27 shows the results for Dataset 1, highlighting that Method 2 outperforms Method 1. Method 2 achieves a higher Silhouette Score of approximately 0.389 and a lower Davies-Bouldin Index of about 0.956. These numbers indicate that Method 2 creates clusters that are better separated and more accurately reflect the data's structure. While Method 1 uses 6 clusters, Method 2 opts for 7 clusters, allowing for a more detailed analysis.

TABLE 28. Results of comparison of method 1 and method 2 in dataset 2.

Method	Method 1 (Authored)	Method 2 (Proposed)
	K-Medoids	
K	7	8
Elbow method	7	7
Silhouette Score	0.23299136450440702	0.2415020454436732
Davies-Bouldin Index	1.3808780186113894	1.2952234255914186

Table 28 presents the results for Dataset 2. Again, Method 2 shows better performance, achieving a Silhouette Score of about 0.242 and a Davies-Bouldin Index of 1.295. These metrics suggest that Method 2 is more effective at forming clear and distinct clusters, even though it increases the number of clusters from 7 (Method 1) to 8 (Method 2).

D. COMPARISON OF CUSTOMER LIFETIME VALUE (CLV) PREDICTION FOR K-MEANS++ AND K-MEDOIDS FOR 2 DATASETS

After a thorough comparison of the methods, method 2 has demonstrated superior results, prompting the selection of K-Means++ and K-Medoids for further exploration in the Customer Lifetime Value (CLV) prediction phase. The subsequent tables present the final CLV predictions for both K-Means++ and K-Medoids applied to the two datasets. These predictions will serve as valuable insights for understanding the long-term value and profitability of customers in the identified clusters.

This subsection focusses on the final CLV predictions derived from the K-Means++ clustering method for Dataset 1 and Dataset 2. The analysis evaluates the accuracy and reliability of the CLV predictions and explores the potential impact on business strategies and customer management initiatives.

The tables 29, 30, 31 and 32 display the values of Recency, Frequency, Monetary, and CLV Prediction for each cluster in both the K-Means++ and K-Medoids methods using two datasets. These results offer valuable insights into strategic decision-making within the company. For instance, clusters with a High CLV Prediction present an opportunity for the company to enhance customer loyalty by offering free gifts to customers within those clusters. On the other hand, clusters with a Low CLV Prediction suggest a potential for customer churn, prompting the company to implement targeted discount and promotion strategies to retain and re-engage customers. Additionally, the results provide the company with an opportunity to observe and analyze customer behavior, enabling informed adjustments to business strategies and marketing efforts. This comprehensive understanding of customer segments allows the company to tailor the approach,

TABLE 29. Results of K-Means++ in dataset 1.

Cluster	Recency	Frequency	Monetary	CLV Prediction
0	Low	High	High	High
1	High	Low	Low	Low
2	Low	Low	Low	Low
3	Low	High	High	High

TABLE 30. Results of K-Means++ in dataset 2.

Cluster	Recency	Frequency	Monetary	CLV Prediction
0	High	High	High	Low
1	High	Low	Low	High
2	Low	High	High	Low
3	Low	Low	Low	High

maximizing customer satisfaction and overall business profitability. Table 29 summarizes the results of the K-Means++ clustering method applied to Dataset 1. Four distinct clusters are identified based on Recency, Frequency, Monetary, and Customer Lifetime Value (CLV) prediction. Cluster 0 is characterized by low recency, high frequency, and high monetary values, indicating a segment of loyal and valuable customers. In contrast, Cluster 1 displays high recency, low frequency, and low monetary values, suggesting a group of less engaged customers. Cluster 2 consists of customers with low recency, low frequency, and low monetary values, representing a segment that may require re-engagement strategies. Finally, Cluster 3 mirrors the characteristics of Cluster 0, with low recency, high frequency, and high monetary values, indicating another group of loyal customers.

Table 30 presents the results of the K-Means++ clustering method applied to Dataset 2. This table also identifies four distinct clusters, but the characteristics of these clusters differ from those in Dataset 1. Cluster 0 features high recency, high frequency, and high monetary values, indicating customers who have recently purchased but may not exhibit long-term loyalty. Cluster 1 shows high recency, low frequency, and low monetary values, suggesting a group of customers who may have made a recent purchase but do not engage regularly. Cluster 2 displays low recency, high frequency, and high monetary values, representing loyal customers, while Cluster 3 has low recency, low frequency, and low monetary values, indicating customers who may need targeted marketing efforts to drive re-engagement.

This discussion centers on the final CLV predictions obtained from the K-Medoids clustering method for Dataset 1 and Dataset 2. The analysis examines the predictive performance of K-Medoids in estimating customer lifetime value

TABLE 31. Results of K-Medoids in dataset 1.

Cluster	Recency	Frequency	Monetary	CLV Prediction
0	Low	High	High	High
1	High	Low	Low	High
2	High	High	High	High
3	High	Low	Low	Low
4	Low	High	High	Low
5	Low	Low	Low	High
6	High	Low	Low	Low

and highlights the implications for strategic planning and resource allocation.

Table 31 summarizes the results of the K-Medoids clustering method applied to Dataset 1. A total of seven distinct clusters are identified based on Recency, Frequency, Monetary values, and Customer Lifetime Value (CLV) prediction. Cluster 0 features low recency, high frequency, and high monetary values, indicating a group of loyal and high-value customers. Cluster 1 shows high recency, low frequency, and low monetary values, suggesting a segment of customers who may not engage frequently but have recently made purchases. Cluster 2 exhibits high recency, high frequency, and high monetary values, pointing to active customers who require retention strategies. Cluster 3 consists of customers characterized by high recency, low frequency, and low monetary values, indicating a need for targeted marketing efforts. Cluster 4 is the same as Cluster 0, with low recency, high frequency, and high monetary values, representing another segment of valuable customers. Cluster 5 includes customers with low recency, low frequency, and low monetary values, suggesting they may be at risk of churn. Finally, Cluster 6 shows high recency, low frequency, and low monetary values, indicating a group that requires immediate attention for engagement.

Table 32 presents the results of the K-Medoids clustering method applied to Dataset 2. Similar to Dataset 1, a total of seven clusters are identified, each with distinct characteristics. Cluster 0 features low recency, low frequency, and high monetary values, suggesting customers who have made significant purchases but may not engage frequently. Cluster 1 displays low recency and high frequency, indicating valuable but less-engaged customers that may require differentiated marketing strategies. Cluster 2 also shows low recency and high frequency, reflecting another segment of loyal customers. Cluster 3 demonstrates high recency, low frequency, and low monetary values, representing customers who have recently interacted but contribute minimally to revenue. Cluster 4 presents high recency, high frequency, and high monetary values, indicating highly engaged customers who may need continuous engagement to maintain loyalty.

TABLE 32. Results of K-Medoids in dataset 2.

Cluster	Recency	Frequency	Monetary	CLV Prediction
0	Low	Low	High	High
1	Low	High	High	Low
2	Low	High	High	Low
3	High	Low	Low	High
4	High	High	High	Low
5	Low	High	Low	High
6	High	Low	Low	Low
7	Low	Low	Low	High

Cluster 5 showcases low recency, high frequency, and low monetary values, while Cluster 6 displays high recency, low frequency, and low monetary values, suggesting both groups require urgent re-engagement efforts.

V. CONCLUSION

In summary, this study on consumer segmentation strategies in online marketing concludes with the aim of uncovering the advantages of utilizing machine learning, particularly unsupervised clustering, to enhance consumer understanding and boost marketing effectiveness.

The exploration begins by closely examining the challenges faced by businesses in understanding diverse customer preferences in the fiercely competitive realm of e-marketing. The absence of a systematic consumer segmentation plan underscores the need for an effective strategy, setting the stage for the research objectives. The outlined study goals provide a clear roadmap, emphasizing the development of a comprehensive understanding of consumer characteristics, a comparative analysis of clustering methods, and the enhancement of marketing effectiveness.

Section II Literature Review delves into various research materials, laying a robust foundation through an in-depth study of customer segmentation and various clustering methods, focusing on the advantages and disadvantages relevant to this study. Section III details the methodology, emphasizing critical steps like data preprocessing, RFM analysis and clustering, which are essential for accurate customer lifetime value prediction. Section IV delivers a critical evaluation of clustering methods, comparing the effectiveness in segmentation and examining the practical implications of customer lifetime value predictions.

The significance of this study lies in the ability to obtain CLV predictions for each cluster through the evaluation results in Section IV. This empowers companies to use CLV predictions for more informed business planning. Additionally, the comparison of clustering methods reveals that K-Means++ and K-Medoids are more suitable for the datasets in this study. Practical takeaways for businesses

include leveraging K-Means++ or K-Medoids for improved segmentation, enabling more accurate targeting and tailored marketing strategies. However, acknowledging the limitations of this research is crucial. The focus was solely on K-Means, K-Means++, K-Medoids, GMM, and Mean Shift clustering methods, limiting the exploration of a broader range of clustering techniques. Future research could address these limitations by incorporating additional clustering methods, exploring scalability issues, and investigating sensitivity to initial parameters or data preprocessing quality.

In comparison to previous studies, the superior performance of K-Means++ consistently demonstrates strong clustering performance and reduced sensitivity to initialization. Overall, this research study contributes to the evolving field of online marketing by demonstrating the advantages of machine learning in consumer segmentation and offering practical insights for marketers to enhance strategies and drive business success.

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